NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT_*COSC(MODEL-A&APEX) | GTA-1(P2) | Date: 22-03-26
Time: 3 Hrs | 2020_P2 | Max. Marks: 198

KEY SHEET
MATHS

1	**5**	2	**4**	3	**3**	4	**8**	5	**6**
6	**3**	7	**BCD**	8	**ABCD**	9	**ABC**	10	**ABD**
11	**AC**	12	**BC**	13	**0.60**	14	**0.50**	15	**0.02**
16	**1**	17	**23**	18	**1**				

PHYSICS

19	**8**	20	**6**	21	**7**	22	**3**	23	**7**
24	**5**	25	**AD**	26	**ACD**	27	**ACD**	28	**ABD**
29	**ACD**	30	**AC**	31	**6.39 TO 6.43**	32	**2.40**	33	**0.45**
34	**8.48**	35	**7.35**	36	**2.57**				

CHEMISTRY

37	**6**	38	**4**	39	**3**	40	**7**	41	**4**
42	**6**	43	**ABC**	44	**CD**	45	**ABD**	46	**ABCD**
47	**ABD**	48	**CD**	49	**4**	50	**25**	51	**23**
52	**23.20 TO 23.50**	53	**30**	54	**44**				

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SOLUTIONS
MATHS

1. Given expression is

$$\left(f(x)f'(x)-x\right)\left(f\left(f'(x)\right)^2+f(x)f''(x)-1\right)=0$$

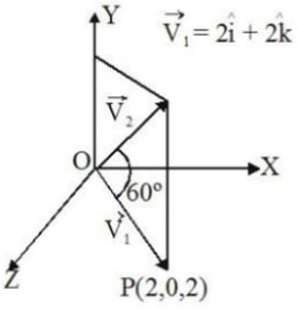
Consider $g(x)=f^2(x)-x^2$

Since $g(x)=0$ has 3 real roots

$\Rightarrow g'(x)=0$ must have minimum 3 roots (including $x=0$)

And $g''(x)=0$ must have minimum 2 real root

2. Equation of plane perpendicular to XOZ pane is $P: x-z=0$, $\vec{V}_2$ is coplanar with $\vec{V}_1$ and



j.

$$\Rightarrow \vec{V}_2=x\vec{V}_1+y\hat{j}$$

$$\Rightarrow \vec{V}_2=x\left(2\hat{i}+2\hat{k}\right)+y\hat{j}$$

$$\therefore \left|\vec{V}_2\right|=\left|\vec{V}_1\right|=2\sqrt{2}$$

$$\therefore 4x^2+y^2+4x^2=8 \qquad \ldots.(1)$$

Now, $\vec{V}_2.\vec{V}_1=\left|\vec{V}_2\right|\left|\vec{V}_1\right|\cos\frac{\pi}{3}=4$

$$4=\left(x\left(2\hat{i}+2\hat{k}\right)+y\hat{j}\right).\left(2\hat{i}+2\hat{k}\right)$$

$$4=8x\Rightarrow x=\frac{1}{2}\Rightarrow y=\pm\sqrt{6}$$

$$\Rightarrow \vec{V}_2=\hat{i}+\sqrt{6}\hat{j}+\hat{k}\left(\because \vec{V}_2.\hat{j}>0\right)$$

Now, $\vec{V}_3=\vec{V}_1+x\hat{i}$

$$\vec{V}_3.\vec{V}_1=0\Rightarrow 8+2x=0\Rightarrow x=-4$$

$$\Rightarrow \vec{V}_3=-2\hat{i}+2\hat{k}$$

$\Rightarrow$ Volume of tetrahedron

$$V=\left|\frac{1}{6}\begin{vmatrix}2 & 0 & 2\\ 1 & \sqrt{6} & 1\\ -2 & 0 & 2\end{vmatrix}\right|$$

$$\Rightarrow V=\frac{1}{6}8\sqrt{6}\Rightarrow\frac{3V}{\sqrt{6}}=4$$

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3. $h'(x)=f(x)$

$h''(x)=f'(x)$

$h(1)=0, f(1)=f'(1)=h'(1)=h''(1)=1=g(1)$

$\Rightarrow f'(g(x))g'(x)=1$

$\Rightarrow f'(g(1))g'(1)=1 \Rightarrow f'(1)g'(1)=1 \Rightarrow g'(1)=1$

$G(x)=x^2g(x)-xh(g(x))$

$G'(x)=2xg(x)+x^2g'(x)-h(g(x))-xh'(g(x))g'(x)$

$=2xg(x)+x^2g'(x)-h(g(x))-x^2g'(x)=2xg(x)-h(g(x))$

$G''(x)=2g(x)+2xg'(x)-h'(g(x))g'(x)=2g(x)+xg'(x)$

$\therefore G''(1)=2g(1)+g'(1)=2+1=3$

4. $f(x)=x^2+\int_0^x e^{-t}f(x-t)dt=x^2+e^{-x}\int_0^x e^t f(t)dt$

$\Rightarrow f'(x)=2x-e^{-x}\left(e\left(f(x)-x^2\right)\right)+e^{-x}.e^x f(x)$

$\Rightarrow f'(x)=2x+x^2 \Rightarrow f(x)=\frac{x^3}{3}+x^2+K$

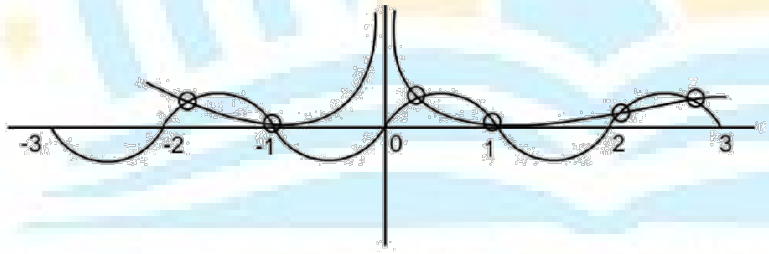
But, $f(0)=0 \Rightarrow k=0$

$\therefore f(1)=\frac{4}{3}$

5. No. of sol of $\sin\pi x=\|n\|x\|\|$

$y=\sin\pi x, y=\|n\|x\|\|$

No of sol=6



No. of solutions 6

6. $(20\sin x)^3+(15\cos x)^2+12^3=3.20\sin x.15\cos x.12$

$(20\sin x+15\cos x+12)$

$\frac{1}{2}.\left[(20\sin x-15\cos x)^2+(15\cos x-12)^2+(12-20\sin x)^2\right]=0$

$\Rightarrow 20\sin x+15\cos x+12=0$ or $20\sin x=15\cos x=12$

Case-I: When $20\sin x+15\cos x+12=0$

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$\Rightarrow 20.\frac{2t}{1+t^2}+15.\frac{1-t^2}{1+t^2}+12=0$

Where $t=\tan\frac{x}{2}$

$\Rightarrow 3t^2-40t-27=0\Rightarrow \tan\frac{x}{2}=\frac{40\pm\sqrt{1924}}{6}$

As $0<x<2\pi$ $\quad\therefore o<\frac{x}{2}<\pi$

$\therefore$ only one solution each in $\left(0,\frac{\pi}{2}\right)\&\left(\frac{\pi}{2},\pi\right)$

Case-II: When $20\sin x=15\cos x=12$

$\Rightarrow \sin x=\frac{3}{5}$ and $\cos x=\frac{4}{5}$

$\therefore$ only one solution in $(0,2\pi)$

$\therefore$ total 3 distinct solutions in $(0,2\pi)$

7. $p(n)=$ probability that child stops after drawing exactly n marbles

$p(n)=\frac{1}{2}.\frac{2}{3}.\frac{3}{4}.....\left(\frac{n-1}{n}\right)\left(\frac{1}{n+1}\right)=\frac{1}{n(n+1)}$

8. $x=\frac{12}{15}\times\frac{1}{\left(1+\frac{1}{x^2+y^2}\right)}$ .....(i)

and $y=\frac{4}{5}\times\left(\frac{1}{1-\frac{1}{x^2+y^2}}\right)$ .....(ii)

Put $x^2+y^2=\frac{1}{u}$ and add $(i)^2$ and $(ii)^2$

$\Rightarrow \frac{1}{u}=\left(\frac{4}{5}\right)^2\left(\frac{9}{(1+4)^2}+\frac{1}{(1-4)^2}\right)$

$\Rightarrow 25\left(u^2-2+\frac{1}{u^2}\right)=32\left\{5\left(4+\frac{1}{4}\right)-8\right\}$

Put $u+\frac{1}{u}=v\Rightarrow v=\frac{6}{5}$ or $\frac{26}{5}$

$\Rightarrow 4=5$ or $\frac{1}{5}$

$\Rightarrow (x,y)=(2,1)$ or $\left(\frac{2}{5},\frac{-1}{5}\right)$

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9. $$\sum_{1\le i<j\le 7}\left|a_i+a_j\right|^2=\sum_{1\le i<j\le 7}\left(\left|a_i\right|^2+\left|a_j\right|^2+a_i\bar{a}_j+\bar{a}_i a_j\right)$$

$$=2R^2.{}^7C_2+\sum_{i\neq j}a_i\bar{a}_j=42R^2+\sum_{i=1}^{7}\sum_{j=1}^{7}a_i\bar{a}_j-\sum_{i=1}^{7}a_i\bar{a}_j$$

$$=35R^2+\left(\sum_{i=1}^{7}a_i\right)\left(\sum_{i=1}^{7}\bar{a}_i\right)-7R^2$$

$$=35R^2+\left|\sum_{i=1}^{7}a_i\right|^2\ge 35R^2$$

10. $$\frac{a}{1+\alpha^2\beta^2}=\frac{b}{\alpha^2+\beta^2}=\frac{c}{\left(1-\alpha^2\right)\left(1+\beta^2\right)}=\frac{2s}{s+2\beta^2}$$

$$\Rightarrow\frac{s}{1+\beta^2}=\frac{a}{1+\alpha^2\beta^2}\Rightarrow\frac{s-a}{\beta^2\left(1-\alpha^2\right)}=\frac{a}{1+\alpha^2\beta^2}$$

$$\Rightarrow s-a=\frac{a\left(1-\alpha^2\right)\beta^2}{1+\alpha^2\beta^2}$$

Similarly $s-b=\dfrac{b\left(1-\alpha^2\right)}{\alpha^2+\beta^2}$

$$s-c=\frac{c\alpha^2}{1-\alpha^2}\ \&\ s=\frac{b\left(1+\beta^2\right)}{\alpha^2+\beta^2}$$

$$\therefore\tan^2\left(\frac{A}{2}\right)=\frac{(s-b)(s-c)}{s(s-a)}=\frac{\dfrac{b\left(1-\alpha^2\right)}{\alpha^2+\beta^2}\cdot\dfrac{c\alpha^2}{1-\alpha^2}}{\dfrac{b\left(1+\beta^2\right)}{\alpha^2+\beta^2}\dfrac{a\left(1-\alpha^2\right)\beta^2}{1+\alpha^2\beta^2}}$$

$$\left[Qa=\frac{c\left(1+\alpha^2\beta^2\right)}{\left(1-\alpha^2\right)\left(1+\beta^2\right)}\right]$$

$$=\frac{\alpha^2}{\beta^2}$$

$$A=2\tan^{-1}\left(\frac{\alpha}{\beta}\right)$$

$$B=2\tan^{-1}(\alpha\beta)$$

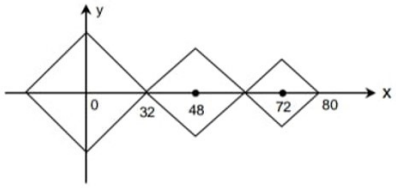
$$\therefore\text{ Required area }=\frac{1}{2}bc\sin A=\frac{1}{2}b.c\frac{2\tan\left(\dfrac{A}{2}\right)}{1+\tan^2\left(\dfrac{A}{2}\right)}=\frac{bc\alpha\beta}{\alpha^2+\beta^2}$$

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11. $a_2 = a_1 + \frac{3}{2}b_1 = 48, b_2 = \frac{b_1}{2} = 16$

$a_3 = 48 + \frac{3}{2}\times 16 = 72, b_3 = \frac{16}{2} = 8$

So, the three loops from $i = 1$ to $i = 3$ are alike Now, area of ith loop (square) $= \frac{1}{2}(\text{diagonal})^2$



$A_i = \frac{1}{2}(2b_i)^2 = 2b_i^2$

$\therefore \frac{A_{i+1}}{A_i} = \frac{2(b_{i+1})^2}{2(b_i)^2} = \frac{1}{4}$

So, the areas from a G.P series

$\therefore$ sum of the G.P up to infinite term is

$\frac{A_i}{1-r} = 2(32)^2 \times 1\frac{1}{1-\frac{1}{4}} = \frac{8}{3}(32)^2$ sq.units

12. $|\bar{a}| = 1, |\bar{b}| = 4, \bar{a}.\bar{b} = 2$ & $\bar{c} + 3\bar{b} = 2\bar{a} \times \bar{b}$

$Q\ \bar{a}.\bar{b} = 2$

$\Rightarrow |\bar{a}||\bar{b}|\cos\theta = 2$

$\Rightarrow \cos\theta = \frac{2}{|\bar{a}||\bar{b}|} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$

Again $|\bar{c} + 3\bar{b}|^2 = |2\bar{a} \times \bar{b}|^2$

$\Rightarrow |\bar{c}|^2 + 9|\bar{b}|^2 + 2\bar{c}.3\bar{b} = 4|\bar{a}|^2|\bar{b}|^2 \sin^2\theta$

$\Rightarrow |\bar{c}|^2 + 144 + 6\bar{b}.\bar{c} = 48$

Also, $\bar{c} = 2\bar{a} \times \bar{b} - 3\bar{b}$

$\Rightarrow \bar{b}.\bar{c} = \bar{b}.(2\bar{a} \times \bar{b}) - 3|\bar{b}|^2 \Rightarrow b.c = 0 - 3 \times 16 = -48$

$\therefore |\bar{c}|^2 + 96 - 6 \times 48 = 0$

$\Rightarrow |\bar{c}|^2 = 192 \Rightarrow |\bar{c}| = 8\sqrt{3}$

$\bar{b}.\bar{c} = |\bar{b}||\bar{c}|\cos\alpha \Rightarrow -48 = 4 \times 8\sqrt{3}\cos\alpha$

$\Rightarrow \cos\alpha = -\frac{6}{4\sqrt{3}} = -\frac{\sqrt{3}}{2}$

$\Rightarrow \alpha = \frac{5\pi}{6}$

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13. Probability that (i) Dice shows 2 and A reports it correctly, $\frac{1}{6}\times\frac{1}{3}=\frac{1}{18}$

(ii) Dice did not show 2 and A reported it as $2,\frac{5}{6}\times\frac{2}{3}\times\frac{1}{5}=\frac{1}{9}$

(iii) Dice showed 4 or 6 and A reported it as $2,\frac{2}{6}\times\frac{2}{3}\times\frac{1}{5}=\frac{2}{45}$

$$P=\frac{\frac{1}{18}+\frac{2}{45}}{\frac{1}{18}+\frac{1}{9}}=\frac{3}{5}$$

14. $$\frac{1}{{}^nC_0}+\frac{1}{{}^nC_n}\le a_n\le\left(\frac{1}{{}^nC_0}+\frac{1}{{}^nC_n}\right)+\left(\frac{1}{{}^nC_1}+\frac{1}{{}^nC_{n-1}}\right)+\sum_{i=2}^{n-2}\frac{1}{{}^nC_2}$$

15. $$T_k=\frac{51!}{50}\left[\frac{1}{(k+1).....(k+50)}-\frac{1}{(k+2)(k+3).....(k+51)}\right]$$

$$S=\frac{1}{50}$$

16. $$\left(ae^{-iB}+be^{iA}\right)^3=\left(c+i\left(b\sin A-a\sin B\right)\right)^3$$

17. $$P(x)=Q(x)(x-1)(x-2)+2x+3$$

$$P(10)=72n+23$$

18. From similar ΔAEM and ΔAPE

$$\frac{AE}{AP}=\frac{AM}{AE}$$

$$\Rightarrow AE^2=AM.AP=(AD-r)AP \qquad ....(1)$$

AB both circles touches each other $(R-r)^2=r^2+OD^2=r^2+(AD-R)^2$

$$AE=AD$$

PHYSICS

19. For equilibrium about the hinge,

$$\pi R\lambda g\times\frac{2R}{\pi}=\lambda Lg\times\frac{L}{2}$$

$$\Rightarrow L=2R$$

$$X_{cm}=\frac{\lambda L\left(-\frac{L}{2}\right)+\pi R\lambda\left(\frac{2R}{\pi}\right)}{\lambda L+\pi R\lambda}=\frac{\frac{L^2}{2}-\frac{L^2}{2}}{\frac{L}{2}[\pi+2]}=0$$

$$Y_{cm}=\frac{\lambda L(L)+\pi R\lambda(R)}{\lambda L+\pi R\lambda}=\frac{L^2+\frac{\pi L^2}{4}}{\frac{L}{2}(\pi+2)}=\frac{L}{2}\left(\frac{\pi+4}{\pi+2}\right)\Rightarrow\alpha=4,\beta=2\Rightarrow\alpha\beta=8$$

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20. Heat lost in a time dt, when the temperature of the gas is T, is $dQ=-\dfrac{K\left(6a^2\right)\left(T-T_0\right)}{x}dt$

Also, volume of the gas (equal to that of the container) remains constant, we have

$dQ=nC_vdT$

$=\dfrac{nRdT}{\gamma-1}$

Thus, $\int_{T_1}^{T}\dfrac{dT}{T-T_0}=-\dfrac{6Ka^2\left(\gamma-1\right)}{nRx}\int_0^t dt$

On solving,

$T=T_0+\left(T-T_0\right)\exp\left[-\dfrac{6ka^2\left(\gamma-1\right)t}{nRx}\right]$

$\Rightarrow p=1;\ q=6;\ s=-1$

$\Rightarrow p+q+s=6$

21. $P_{out}=\dfrac{4S}{2R}=\dfrac{S_{out}RT}{M}$

$P_{in}=\dfrac{4S}{R}+\dfrac{4S}{2R}=\dfrac{6S}{R}=\dfrac{S_{in}RT}{M}$

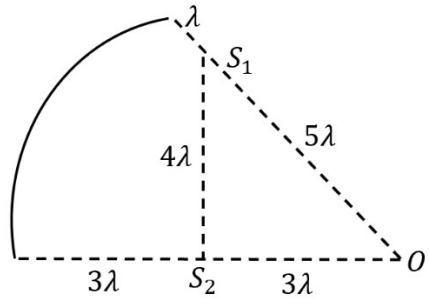
$\Rightarrow\dfrac{S_{in}}{S_{out}}=\dfrac{P_{in}}{P_{out}}=3$

Temperature of the gases in the bubbles are the same

$\Rightarrow\dfrac{P_{out}V_{out}}{n_{out}R}=\dfrac{P_{in}V_{in}}{n_{in}R}\Rightarrow\dfrac{n_{out}}{n_{in}}=\dfrac{\dfrac{2S}{R}}{\dfrac{6S}{R}}\times\dfrac{\dfrac{4}{3}\pi\left(TR^3\right)}{\dfrac{4}{3}\pi R^3}=\dfrac{7}{3}=\dfrac{1}{7}$

Thus, $\dfrac{X}{Y}=7$

22.



At the slits, the path difference between the two arriving loads is 2λ.

From the slits to E the path difference between the waves is $\Delta x=S_1E+2\lambda-S_2E-\left(\mu-1\right)t$

$=\left(S_1E-S_2E\right)+2\lambda-\dfrac{t}{2}$

For CM to be at $E,\Delta x=0$

$\Rightarrow\dfrac{t}{2}=\left(S_1E-S_2E\right)+2\lambda$

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Now, $S_2E^2 - S_1E^2 = (4\lambda)^2$

$2S_1E(S_2E - S_1E) = 16\lambda^2$

$\Rightarrow S_1E - S_2E = -\frac{\lambda}{16}$

$\Rightarrow \frac{t}{2} = 2\lambda - \frac{\lambda}{16} \rightarrow t = \frac{31\lambda}{8} \Rightarrow k = 3$

23. Let $\int_P^Q \vec{B}_A.\vec{dl} = -9\mu_0 = x$ and $\int_R^S \vec{B}_B.\vec{dl} = 5\mu_0 = y$

Now, $\oint \vec{B}.\vec{dl} = \mu_0(-i_0 + 2i_0 + 8i_0 - 4i_0) = 5\mu_0 i_0$

Since B-field αi, from symmetry

$\oint \vec{B}.\vec{dl} = \left(2x - \frac{2y}{2}\right)_A + (-4x + 2y)_B + (-16x + 8y)_C + (8x - 4y)$

$5\mu_0 i_0 = -10x + 5y$

$= -10(-9\mu_0) + 5(5\mu_0)$

$i_0 = 23$

$= 3(7) + 2$

$\Rightarrow p = 7$

24. The potential due to an element of the rod at distance y from B is,

$dV = -\frac{G\left(\frac{M}{L}\right)dx}{y + x}$

$V = -\frac{GM}{L}\int_0^L \frac{dx}{y + x}$

$V = -\frac{GM}{L}\ln\left(1 + \frac{L}{y}\right)$

$y = L \Rightarrow V_i = -\frac{GM}{L}\ln(2)$

$y = \frac{L}{2} \Rightarrow V_f = -\frac{GM}{L}\ln(3)$

Using conservation of energy,

$0 + mV_i = \frac{1}{2}mu^2 + mv_f$

$\Rightarrow \frac{u^2}{2} = \frac{GM}{L}[\ln(3) - \ln(2)]$

$u = \sqrt{\frac{2GM}{L}\ln\left(\frac{3}{2}\right)}$

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25. The two masses execute circular motion about their center of mass

$$V_{CM}=\frac{mV_0+M(0)}{m+M}$$

$$V_{1CM}=V_0-\frac{mV_0}{m+M}=\frac{MV_0}{m+M}$$

$$V_{2CM}=0-\frac{mV_0}{m+M}=-\frac{mV_0}{m+M}$$

Also, $\frac{\left(\frac{MV_0}{M+m}\right)^2}{\frac{MR}{M+m}}=\frac{V_0^2}{R_C}\Rightarrow R_C=\frac{(M+m)R}{M}$

Thus, $N=\frac{mV_0^2}{R_C}=\frac{mM}{M+m}\times\frac{V_0^2}{R}$

If, $M=2m, N=\frac{2mV_0^2}{3R}$

26. First resonance occurs when, $1000=\frac{320}{4l_1}\Rightarrow l_1=8cm$

Second resonance occurs when, $1000=\frac{3\times320}{4l_2}\Rightarrow l_2=24cm$

The time at which the height reduces to h_1 from h_2 is

$$t=\frac{2A}{a}\times\frac{1}{\sqrt{2g}}\left[\sqrt{h_1}-\sqrt{h_2}\right]$$

For first resonance, $h_1=100cm$, $h_2=100-8=92cm$

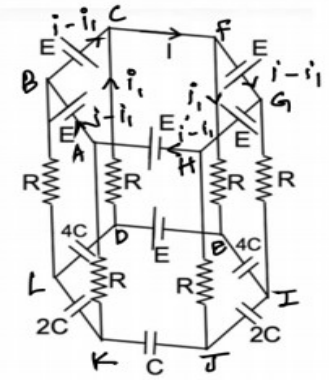
$$t_1=\frac{2\times2000}{1}\times\frac{1}{\sqrt{20}}\times\left[1-\sqrt{\frac{92}{100}}\right]=35.7s\approx0.6\text{ min}$$

For second resonance, $h=100-24=76\ cm$

$$t_2=\frac{2\times2000}{1}\times\frac{1}{\sqrt{20}}\left[1-\sqrt{\frac{76}{100}}\right]=116.3s=1.93\text{min}$$

$$\Delta t=t_2-t_1\approx1.3\text{min}$$

27. During steady state,



KVL to *ABCDEFGHA*

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$4E-5(i-i_1)r+i_1r+2i_1R$

$20-5i+10i_1=0$

KVL to $CFEDC$

$E-2i_1R-i_1r=0$

$5-5i_1=0\Rightarrow i_1=1$

$\Rightarrow 5i=30\Rightarrow i=6A$

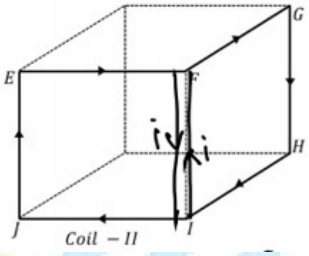
KVL to $FGIEF$, (q_1 is charge on $4C$)

$$5-5(1)+\frac{q_1}{8\times10^{-6}}+2(1)=0\Rightarrow q_1=16\mu C$$

KVL to $AHJKA$, (q_2 is charge on C)

$$-5+5+\frac{q_2}{2\times10^{-6}}=0\Rightarrow q_2=0.$$

28. By definition, $\phi_I=L_1i$ and $\phi_{II}=L_2i$



Now, adding and subtracting i as shown and if ϕ_n is the flux through the neighboring surface, then

$\phi_{EFIJ}=L_1i+\phi_n$

$\phi_{FGHI}=L_1i+\phi_n$

Thus,

$\phi_{II}=\phi_{EFIJ}+\phi_{FGHI}=2L_1i+2\phi_n$

$$\Rightarrow L_2i=2L_1i+2\phi_n\Rightarrow\phi_n\left(\frac{L_2}{2}-L_1\right)i.$$

29. $V_1^2=V_L^2+V_R^2=\left(50\sqrt{2}\right)^2$

$V_2=V_L\sim V_C=120$

Also, $V_R^2+\left(V_L-V_C\right)^2=130^2$

$V_R^2=130^2-120^2$

$V_R=50V$

$\Rightarrow V_L^2=5000-2500$

$\Rightarrow V_L=50V$

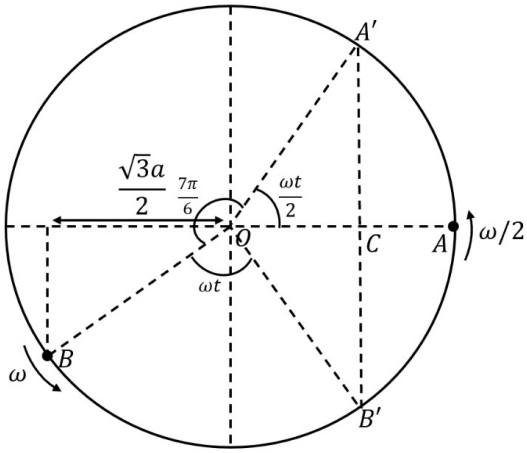
$\Rightarrow V_L=170V$

$V_C>V_L\Rightarrow$ capacitive

Also $\cos\phi=\frac{R}{Z}=\frac{V_R}{V_{net}}=\frac{50}{130}=\frac{5}{13}$

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30. Motion of m is the projection of B on the $x-$axis and motion of $4m$ is the projection of A on the $x-$axis. The collision occurs for the first time when B travels to B' and A travels to A'. A and B rotate with speeds $\frac{\omega}{2}$ and ω respectively, where $\omega=\sqrt{\frac{m}{k}}$.



When they collide at a time t, then

$$\frac{7\pi}{6}+\omega t=2\pi-\frac{\omega}{2}t\Rightarrow\omega t=\frac{5\pi}{9}.$$

Thus, $t=\frac{5\pi}{9}\sqrt{\frac{m}{k}}$. Also, the distance $OC=a\cos\left(\frac{\omega t}{2}\right)=a\cos\left(\frac{5\pi}{18}\right)$.

31. Since $\mu\sin\theta=$ constant

$$2\sin 60^\circ=\frac{1}{1+H^2}\sin 90^\circ$$

$$\Rightarrow 1+H^2=\frac{2}{\sqrt{3}}$$

$$H^2=\frac{2}{\sqrt{3}}-1$$

$$\frac{1}{H^2}=\frac{1.73}{2-1.73}$$

$$=6.41$$

32. $d\phi=(1)\left(\frac{\mu_0 i}{2\pi x}\right)(4dx)$

$$\phi=\frac{2\mu_0 i}{\pi}\int_2^4\frac{dx}{x}$$

$$\phi=\frac{2\mu_0 i}{\pi}\ln(2)$$

$$=\frac{2\mu_0 i_0}{\pi}\ln(2)\sin\left(\omega t+\frac{\pi}{6}\right)$$

$$\varepsilon=\frac{2\mu_0 i_0\omega}{\pi}\ln(2)\cos\left(\omega t+\frac{\pi}{6}\right)\text{ (in magnitude)}$$

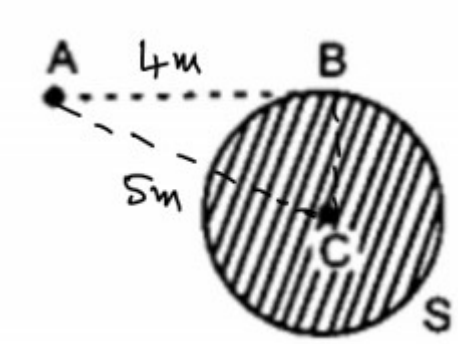
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$$E|_{t=0} = E_0 \frac{\sqrt{3}\mu_0 i_0 \omega \ln(2)}{\pi} \Rightarrow q|_{t=0} = \frac{\sqrt{3}\mu_0 i_0 \omega C \ln(2)}{\pi}$$

$$q|_{t=0} = \sqrt{3} \times 4 \times 10^{-7} \times 10^6 \times 5 \times 0.693 \ \mu C$$

$$= 2.40 \ \mu C$$

33.



$$V_C^{tot} = V_q^C + C_{ind}^C \left(V_{ind} = 0\right)$$

$$= 9 \times 10^9 \times \frac{10^{-6}}{5}$$

$$= 1.8 kV$$

$V_B^{tot} = V_C^{tot}$ $\because$ of conducting nature, the entire volume is equipotential.

$$1.8 \ kV = V_q^B + V_{ind}^B$$

$$\Rightarrow V_{ind}^B = 1.8kV - 9 \times 10^9 \times \frac{10^{-6}}{4}$$

$$= -0.45kV$$

34. $n \ VSD = (n-4) MSD$

$$1VSD = \frac{n-4}{n} MSD = \frac{n-4}{n} mm$$

$$LC = 0.05 \ mm = \left(1 - \frac{n-4}{n}\right) mm$$

$$\Rightarrow n = 80$$

$$CVD = \frac{80}{5} = 16^{th} \text{ division}$$

$$R = MSR + CVD \times LC$$

$$= 8.4 + 16 \times 0.005$$

$$= 8.48 \ cm$$

35. $P = Nhv$; N : No. of photons emitted per second

$$i_{sat} = 10 \times 10^{-6} \ A$$

$$n \times 1.6 \times 10^{-19} = 10^{-5}$$

$$n = 6.25 \times 10^{13}$$

n : no. of photoelectrons emitted per second.

Given,

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$$n = \frac{10^{-3}}{100} \times N$$

$$\Rightarrow N = 6.25 \times 10^{18}$$

$$h\nu = W + eV_S$$

$$h\nu = 2.35 + 5$$

$$= 7.35 \times 1.6 \times 10^{-19} J$$

$$P = 6.25 \times 10^{18} \times 7.35 \times 1.6 \times 10^{-19}$$

$$= 7.35W$$

36. The forces acting on the dielectric are an attractive electrostatic force of field of capacitor and its weight. The block will slip when

$$F_E + \frac{m}{6}g \geq \mu mg \Rightarrow F_E \geq \frac{mg}{12}$$

Now, $F_E = \frac{\varepsilon_0 l V^2}{2d}(k-1)$

Thus, $V^2 \geq \frac{mgd}{6\varepsilon_0 l(k-1)}$ and

$$V_{\min} = \sqrt{\frac{mgd}{6\varepsilon_0 l(k-1)}} \Rightarrow x = 6.$$

If $V = 2V_{\min} = \sqrt{\frac{2mgd}{3\varepsilon_0 l(k-1)}}$, then

$$F_E = \frac{\varepsilon_0 l}{2d}(k-1) \times \frac{2mgd}{3\varepsilon_0 l(k-1)} = \frac{mg}{3}$$

Also,

$$F_E + \frac{mg}{6} - \frac{mg}{4} = \frac{7m}{6}a \Rightarrow a = \frac{3g}{14}$$

Now,

$$v^2 = 0 + 2 \times \frac{3g}{14} \times (l-b) \Rightarrow v = \sqrt{\frac{3}{7}g(l-b)}$$

Thus, $y = \frac{3}{7} \Rightarrow xy = \frac{18}{7} \approx 2.57$.

CHEMISTRY

37. Three moles of alcohol require four moles of nitric acid for its oxidation to carboxylic acid.

38. $m_{air} = m_{balloon} + m_{gas}$

$V_{balloon} \times M_{air} = 8 + n_g \times M_g$

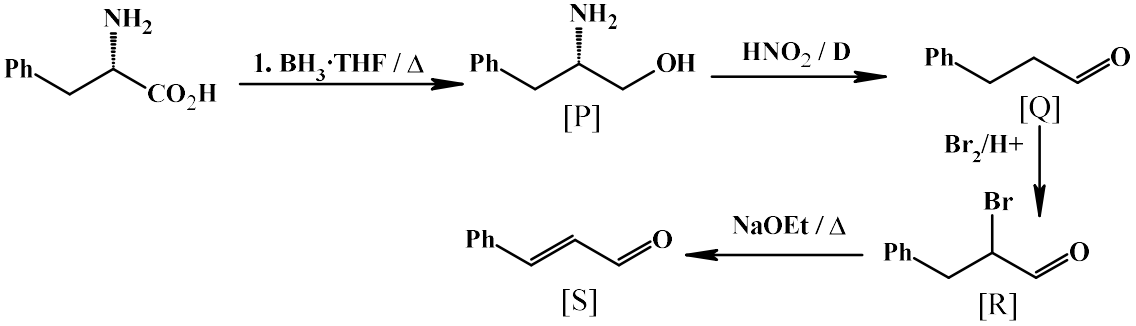
Moles of gas, $n_g = \frac{PV}{RT} = 2$

Thus $M_g = 25$, From this molar ratio of O_2 & He is found to be 75:25

The mass ratio = 96 : 4

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39. ii) and v) are incorrect.



40. $Ecell = E^0_{cell} - 0.06pH$

$0.158 = (0.70 - 0.242) - 0.06pH$

$pH = 5$; Ka = $2x10^{-5}$

41. Only (iv) and (vi) are correct

$P_4 + 8SOCl_2 \rightarrow 4PCl_3 + 4SO_2 + 2S_2Cl_2$

(A) (C) (B)

42. All a, b, c, d, e, f are correct

43. A) $SO_2 + 2H_2O + 2CuSO_4 + 2KSCN \rightarrow K_2SO_4 + 2H_2SO_4 + 2CuSCN$

B) $P_4 + 10HNO_3 + H_2O \rightarrow 4H_3PO_4 + 5NO + 5NO_2$

C) $5HSO_3^- + 2IO_3^- \rightarrow 3HSO_4^- + 2SO_4^{2-} + I_2 + H_2O$

D) $2KNO_3(s) + 10K(s) \rightarrow 6K_2O(s) + N_2$

44. 

45. $-\frac{dC_{reactant}}{dt} = Rate$ & As temperature increases, rate constant of a chemical reaction increases and there by half life time decreases, $t_{1/2}$ of first order reaction is independent of initial concentration of reactant. fraction of reactant remaining after

8 half lifes is = $\frac{1}{2^8}$

% Remaining = $\frac{100}{2^8} = 0.4$

% Reacting = 100 – 0.4 = 99.6%

46. A) Excess of lithium makes LiCl crystals pink due to F – centres.
B) When silicon is doped with group-15 element n-type semiconductor is formed.
C) H_2O, NaCl and C_6H_6 are some examples of diamagnetic substances.
D) AgBr shows both schottky and frenkel defects.

47. In Ln^{3+} size decreases smoothly as atomic number increases.
D and f-orbitals are more diffused and as a result their shielding is poor. Also shielding by a d-electron is greater than that of an f-electron in the same subshell

48. Conceptual

49. Conceptual

50. $\overset{+3}{C_2}O_4^{2-} \longrightarrow \overset{+4}{CO_2}$

n factor of $K_xH(C_2O_4)_y \Rightarrow 2y$ and also 2y = x + 1

meq of salt + meq of Mohr salt = meq of $KMnO_4$

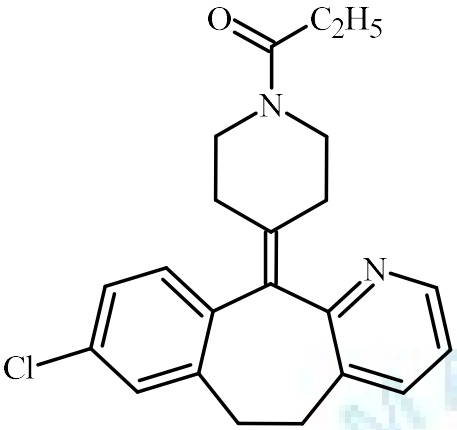
on solving, y =2, since x + 1 = 2y, $\therefore$ x = 3.

Molecular formula of the compound is $K_3H(C_2O_4)_2$

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51. $\text{D.U.} = \dfrac{(2n+2)-(\text{no. of H atoms}+\text{no. of X atoms}-\text{no. of N atoms})}{2}$

$\therefore$ No. of hydrogen atoms = 23



52. $\Delta S^0 = -\left[\dfrac{d(\Delta G^0)}{dT}\right]_P$

$\Delta S^0 = -4\ln T - 0.5 = 24.45 \text{ cal mol}^{-1}$

$\Delta H^0 = \Delta G^0 + T\Delta S^0$

$= -13519 - (400 \times 24.45) = -23.3 \text{ kcal mol}^{-1}$

53. Fe^{2+} in $[Fe(H_2O)_2]^{2+}$ is an octahedral high spin complex with $t_{2g}^4 e_g^2$ configuration

12 Kcal mol^{-1} is the CSFE from which the Δ_0 value can be calculated as follows:

CSFE = $[4(-0.4)\ \Delta_0 + 2(0.6)\ \Delta_0] + 0 = -0.4\Delta_0$

$\therefore \quad \Delta_0 = \dfrac{12}{0.4} = 30 \text{ Kcal mol}^{-1}$

54. $m = \dfrac{\Delta T_f}{K_f} = \dfrac{2.7}{5.12} = 0.527 m$

$$0.527 = \frac{\dfrac{x}{128} + \dfrac{1.6-x}{178}}{0.02}$$

On solving $x = 0.7$g

% Naphthalene = 44

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